



Medical Policy

Subject: Near-Infrared Coronary Imaging and Near-Infrared Intravascular Ultrasound Coronary Imaging

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Description/Scope

This document addresses near-infrared coronary imaging and near-infrared intravascular ultrasound coronary imaging. Near-infrared coronary imaging, also referred to as near-infrared spectroscopy (NIRS), was created for the examination of coronary arteries to detect plaques with large lipid cores in individuals undergoing cardiac angiography. Subsequently, near-infrared intravascular ultrasound coronary imaging was developed for the examination of coronary arteries during invasive coronary angiography in combination with intravascular ultrasound (IVUS) examination of coronary intravascular pathology.

Position Statement

Investigational and Not Medically Necessary:

Near-infrared coronary imaging is considered **investigational and not medically necessary** for all indications.

Near-infrared intravascular ultrasound coronary imaging is considered **investigational and not medically necessary** for all indications.

Rationale

Summary

Although near-infrared coronary imaging and intravascular ultrasound coronary imaging, particularly the LipiScan Coronary Imaging System and the LipiScan IVUS Imaging System (Infraredx, Inc., Burlington, MA), were FDA cleared for detecting lipid-core plaques in 2008 and 2010 respectively, their clinical utility remains unclear. In addition, dual-modality NIRS + IVUS imaging systems from the same manufacturer have been FDA cleared between 2013 and 2025. Despite studies indicating a correlation between the lipid core burden index (LCBI) and major adverse cardiovascular events, significant methodological limitations of those studies have been identified and the need for larger confirmatory studies persists. Technical issues such as difficulty in determining plaque depth, susceptibility to influences of temperature and pH, and insufficient evidence from peer-reviewed literature further limit the acceptance of these imaging technologies in enhancing health outcomes.

Discussion

Near-infrared coronary imaging (LipiScan™ Coronary Imaging System, Infraredx, Inc., Burlington, MA) was cleared for marketing through the U.S. Food and Drug Administration (FDA) 510(k) process in April 2008. According to the 510(k) summary, the device is intended for the near-infrared examination of coronary arteries, for the detection of lipid-core-containing plaques of interest, and for the assessment of coronary artery lipid core burden. The LipiScan Coronary Imaging System uses the same basic catheter design and the same operating principle as the predicate device, the Infraredx NIR System (Infraredx, Inc., Burlington, MA) which was cleared for marketing by the FDA 510(k) process in June 2006 and is also intended for the near-infrared imaging of coronary arteries.

Near-infrared intravascular ultrasound coronary imaging (Infraredx LipiScan IVUS Imaging System, Infraredx, Inc., Burlington, MA) was cleared for marketing by the FDA 510(k) process in June 2010. This system combines both near-infrared and intravascular ultrasound technologies. According to the 510(k) summary, modifications from the LipiScan Coronary Imaging System to the LipiScan IVUS Imaging System are the inclusion of ultrasound imaging within the same dimensions of the catheter and an expanded indication for use (ultrasound examination of coronary intravascular pathology). The Infraredx LipiScan IVUS imaging system uses the same basic catheter design and the same operating principle as the predicate LipiScan Coronary Imaging system, while the ultrasound capabilities are functionally equivalent to the iLab™ Ultrasound Imaging System (Boston Scientific Corp., Fremont, CA).

The TVC Imaging System™ (Infraredx, Inc., Burlington, MA), also a dual-modality NIRS/IVUS imaging device, was cleared for marketing by the FDA 510(k) process in March 2013. The Makoto™ Intravascular Imaging System (Infraredx, Inc., Bedford, MA), which uses the Dualpro™ IVUS + NIRS catheter, is currently available in the U.S. It was cleared for marketing by the FDA 510(k) process in April 2019 based on the Lipid-Rich Plaque (LRP) study (NCT02033694). Similar Makoto Intravascular Imaging Systems which also include a peripheral imaging catheter received 510(k) clearance from the FDA in December 2021 and in February 2025.

Several studies published before 2015 examined the capabilities and limitations of NIRS in detecting lipid core plaques (LCPs) and predicting coronary events. Gardner (2008) validated NIRS using autopsy specimens, finding a 4.3% presence of LCPs but noted limitations including the use of non-living tissue and small sample size. Studies by Garcia-Garcia (2008) and Schaar (2007) explored in vivo NIRS feasibility, concluding that while techniques were promising, they were unproven and susceptible to technical challenges such as blood scattering and pH effects. Suh (2011) emphasized that NIRS could only detect one feature of vulnerable plaques and could not assess plaque depth. Madder et al. (2013) used combined NIRS and IVUS in individuals with ST segment elevation myocardial infarction (STEMI), showing high accuracy in differentiating culprit lesions with a proposed maximum lipid core burden index in a 4 mm length of artery ($\text{maxLCBI}_{4\text{mm}}$) threshold (> 400), though these findings were not prospectively validated. Similarly, Roleder et al. (2014) used NIRS-IVUS and optical coherence tomography (OCT) to detect thin-cap fibroatheromas (TCFA), identifying combined LCBI and remodeling index thresholds with high diagnostic accuracy, but the small size and single-arm design of this study requires validation from larger controlled studies. Oemrawsingh (2014) conducted a prospective study linking higher LCBI values with worse outcomes over 1 year in individuals with stable and acute coronary syndrome, but did not explore therapeutic impacts of NIRS. Across these studies, a common theme is that NIRS's diagnostic potential is tempered by methodological limitations, small sample sizes, and the need for large-scale prospective validation.

In 2016, Madder and colleagues evaluated NIRS and IVUS findings in pre-existing stents. A comparison was made between NIRS findings in pre-existing stents in which an increased lipid signal possibly indicated neoatherosclerosis and NIRS findings in a control group of newly implanted stents. In the group of pre-existing stents IVUS was used to determine if neointimal tissue was present at the location of lipid-rich plaque (LRP) detected by NIRS. LCBI and maximum LCBI were measured within stented segments. Of 60 pre-existing stents, NIRS detected LRP in 33%. At the location of LRP, IVUS found no neointimal tissue in 35% of cases. NIRS findings in pre-existing stents were reported as indistinguishable from those of the newly implanted stents. The authors concluded that detection of LRP in a pre-existing stent by NIRS alone is not reliable evidence of neoatherosclerosis but that IVUS may provide insight into the potential source of the lipid signal in pre-existing stents. Study limitations included a small sample size and single-center design. Future studies are needed to determine the clinical relevance of NIRS-IVUS findings in pre-existing stents.

Danek and colleagues (2017) performed a retrospective registry study to find factors associated with major adverse cardiovascular events (MACE) during the follow-up of individuals who had NIRS imaging. The primary endpoint was the incidence of MACE defined as the composite of cardiac death, acute coronary syndrome, unplanned coronary revascularization, and stroke. The researchers reviewed the records of 239 participants who had NIRS between 2009 and 2011. They used a Multivariable Cox regression analysis to find variables independently associated with MACE. At a median follow-up of 5.4 years, MACE occurred in 100 participants (41.8%), and 31 MACE events were related to the target vessel. The Kaplan-Meier estimated MACE rate was 37.5% at 5 years. Factors independently associated with MACE were diabetes, prior MI, percutaneous coronary intervention performed at index procedure, and non-target vessel LCBI. The adjusted hazard ratio (HR) for non-target vessel $\text{LCBI} \geq 77$ was 14.05 (95% confidence interval [CI], 2.47 to 133.51; $p=0.002$). The 5-year cumulative incidence of MACE in the non-target LCBI group was 58% compared to 13.1% in the below threshold group. The researchers concluded that "non-target vessel lipid burden measured using NIRS appears to be a predictor of MACE during long-term follow-up." Limitations of the study included the retrospective design, potential selection bias, and small sample size. This study did not investigate whether NIRS could be used to guide treatment in a way that improves health outcomes.

Schuurman and colleagues (2018) conducted a follow-up study to determine the long-term prognostic value of lipid-rich core-containing plaques evaluated by NIRS in individuals with coronary artery disease (CAD). The researchers combined the populations of two previous studies, the ATHEROREMO-NIRS (Oemrawsingh, 2014) and the IBIS-3-NIRS (Netherlands trial register NTR2872). The primary endpoint was MACE, which included all-cause death, non-fatal acute coronary syndrome, and unplanned coronary revascularization during long-term follow-up. Between 2009 and 2013, 275 participants underwent NIRS for acute coronary syndrome or stable angina. In January 2015, the researchers sent follow-up questionnaires to all living participants to identify adverse events. If an event was identified, the researchers reviewed hospital discharge records. Hospital records were also investigated for participants who did not return the questionnaire. At a median of 4.1 years follow-up, 79 participants had MACE. For every $\text{MaxLCBI}_{4\text{mm}}$ increase of 100 units, there was a 19% increase in MACE (HR, 1.19; 95% CI, 1.07 to 1.32; $p=0.001$). The researchers concluded that "LCBI values were significantly and independently associated with the incidence of adverse cardiac outcome in patients with CAD over 4 years of follow-up." Limitations of the study included that the IBIS-3 population received rosuvastatin after the index procedure, there was a 10% loss to follow-up for the questionnaire, and the study had a small sample size. This study did not investigate whether NIRS could be used to guide treatment in a way that improves health outcomes.

The LRP study findings were published in 2019 by Waksman and colleagues. This was a prospective uncontrolled study which enrolled 1563 individuals with known or suspected CAD who underwent cardiac catheterization with possible ad hoc percutaneous coronary intervention. Participants underwent NIRS-intravascular ultrasound imaging with the Infraredx device. All of the individuals with a large LRP (defined as $\text{maxLCBI}_{4\text{mm}} < 250$) detected by NIRS and a randomly selected half of the individuals with a small LRP were contacted for follow-up at 6-month intervals, up to 24 months. The primary outcome was non-culprit major adverse cardiovascular events (NC-MACE). The analysis included 1271 individuals with analyzable data who were followed for up to 24 months. During the follow-up period, NC-MACE was reported in 9% of participants ($n=103$). In a multivariate model controlling for demographic variables and co-morbidities, the $\text{maxLCBI}_{4\text{mm}}$ was significantly associated with NC-MACE. For $\text{maxLCBI}_{4\text{mm}}$ as a continuous variable, the adjusted HR of experiencing MACE for every 100-unit increase, was 1.18 (95% CI, 1.05 to 1.32, $p=0.043$). $\text{maxLCBI}_{4\text{mm}} > 400$ was associated with an HR of 1.89 (95% CI, 1.26 to 2.83; $p=0.0021$). A limitation of the study is that it did not compare outcomes in individuals who were managed with and without use of NIRS-intravascular ultrasound imaging.

In 2022, Omatsu and colleagues conducted a single-center prospective observational study aimed at correlating measurements of yellow plaque obtained by angioscopy with LCBI on NIRS. A total of 95 lesions in 44 participants were analyzed both by NIRS-IVUS followed by coronary angioscopy. Angioscopy results were classified into 4 yellow color grades (0, white; 1, slight yellow; 2, yellow; and 3, intensive yellow) at a location of maximal LCBI on NIRS in each lesion. Angioscopic yellow color grade and LCBI were found to be positively correlated (95% CI 122.9-206.7; $p<0.001$). LCBI was significantly different among different yellow color grades determined by coronary angioscopy (ANOVA, $p<0.001$). The authors concluded that the qualitative angioscopic assessment was objectively validated by the quantitative NIRS evaluation. However, there were some limitations to the study. The sample size was small, limiting the statistical power and impairing the generalizability of the findings. There was also a potential for selection bias since individuals with small or tortuous vessels not suitable for angioscopic examination were excluded. The results of this single-center study might not be generalizable to other settings. This study showed a relationship between angioscopy and NIRS but did not show that NIRS can be used to improve health outcomes.

In 2023, Bass and colleagues performed a systematic review and meta-analysis of the ability of NIRS to identify vulnerable individuals and plaques. A total of seven observational studies with prospective follow-up involving 2948 participants were included in the study. All studies were considered at high overall risk of bias based on the Cochrane Risk of Bias tool. In the pooled meta-analysis, identification of vulnerable plaques with NIRS was associated with 2.93 times increased odds of major adverse cardiovascular or cerebrovascular events (95% CI, 1.82-4.73, $I^2 = 58.7\%$). The authors concluded that NIRS is an effective tool for identifying vulnerable individuals and plaques. However, limitations include the observational nature of the studies, high risk of bias, relatively short-term follow-up (most of the included studies were less than 4 years), and the fact that the primary outcomes for each study included a range of LCBI value thresholds to determine the odds ratio. The authors acknowledged that a definitive optimal LCBI threshold for prediction of subsequent adverse events has yet to be determined.

Currently, there is insufficient evidence available in the peer-reviewed, published literature to determine the clinical utility of both near-infrared coronary imaging and near-infrared intravascular ultrasound imaging systems. Large, well-designed studies demonstrating the effectiveness of these technologies in improving clinical outcomes are needed.

Background/Overview

Coronary artery plaque is a deposit consisting of cholesterol-rich fat, calcium, and other substances found in the blood. As plaque accumulates on the artery wall, it reduces blood flow to the heart muscle and increases the risk of blood clots, which can lead to a heart attack. Vulnerable plaque is coronary artery plaque that is unstable and at high risk of rupturing, thereby causing a clinical cardiovascular event.

The LipiScan Coronary Imaging System is a device created to detect and evaluate coronary artery plaque. It works by inserting a fiberoptic laser into the artery and measuring the wavelengths that are reflected back from the artery wall. The light reflected back at different wavelengths is analyzed to detect the chemical composition of the coronary plaque. A color-coded map is produced by the device console showing the intensity and location of lipid core within plaques of interest in the artery. A lipid core burden index is also reported, which is a measure of the total amount of lipid core containing plaques of interest in the coronary artery.

The InfraReDx LipiScan IVUS Imaging System has a similar intended use and functionality as the LipiScan Coronary Imaging System and also includes intravascular ultrasound imaging of coronary intravascular pathology. A catheter accesses the coronary vasculature and the device output is an image of the artery wall, as an adjunct to coronary angiography.

In a similar manner to the InfraReDx LipiScan IVUS, the TVC Imaging System uses integrated near-infrared and intravascular ultrasound technology to identify lipid-core plaque and intravascular pathology.

Definitions

Coronary arteries: Blood vessels supplying blood to the heart.

Fibroatheroma: Lipid-rich plaque suspected to be a type of vulnerable plaque.

Lipid: A small molecule soluble in non-polar solvents and insoluble in polar solvents such as water. In medicine, the term lipid commonly refers to triglycerides, fats, and sterols.

Lipid core plaque: an atherosclerotic lesion with a dense accumulation of extracellular lipids occupying an extensive but well-defined region of the intima known as the lipid core.

Lipid core burden index (LCBI): a measure of the total amount of lipid core containing plaque assessed by imaging of the coronary arteries.

Major Adverse Cardiovascular Event (MACE): an overlapping range of adverse cardiovascular-related events that can include atherosclerotic or other cardiovascular death, non-fatal myocardial infarction, acute coronary syndrome, coronary or peripheral revascularization, unstable angina, and stroke, as well as heart failure, re-infarction, non-fatal re-infarction, re-hospitalization for cardiovascular-related illness, repetition of percutaneous coronary intervention, and all-cause mortality.

Near-infrared coronary imaging: also referred to as near-infrared spectroscopy, is an imaging modality for the examination of coronary vessels that uses near-infrared light to measure diffuse reflectance signals to detect coronary lipid-rich lesions in individuals undergoing cardiac angiography.

Near-infrared intravascular ultrasound coronary imaging: an intracoronary catheter-based imaging modality that combines near-infrared spectroscopy with intravascular ultrasound technology to quantify lipid accumulation measured as the Lipid Core Burden Index and associate it with other characteristics such as lumen size and plaque architecture.

Near-infrared spectroscopy (NIRS): also referred to as near-infrared coronary imaging, is an imaging modality for the examination of coronary vessels that uses near-infrared light to measure diffuse reflectance signals to detect coronary lipid-rich lesions in individuals undergoing cardiac angiography.

Neoatherosclerosis: A phenomenon in which lipid-rich plaques develop within pre-existing stents.

Thin-cap fibroatheromas (TCFA): atherosclerotic lesions composed of a lipid-rich core containing many lipid-laden macrophage foam cells derived from blood monocytes with a thin, friable cap.

Coding

The following codes for treatments and procedures applicable to this document are included below for informational purposes. Inclusion or exclusion of a procedure, diagnosis or device code(s) does not constitute or imply member coverage or provider reimbursement policy. Please refer to the member's contract benefits in effect at the time of service to determine coverage or non-coverage of these services as it applies to an individual member.

When services are Investigational and Not Medically Necessary:

CPT

93799	Unlisted cardiovascular service or procedure [when specified as intravascular catheter-based coronary vessel or graft spectroscopy (eg, infrared) during diagnostic evaluation and/or therapeutic intervention including imaging supervision, interpretation and report, each vessel]
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ICD-10 Procedure

8E023DZ	Near infrared spectroscopy of circulatory system, percutaneous approach
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ICD-10 Diagnosis

All diagnoses

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InfraReDx LipiScan NIR Catheter Imaging System
 InfraReDx LipiScan IVUS Imaging System
 LipiScan Coronary Imaging System
 Makoto Intravascular Imaging System
 Near-Infrared Imaging (NIR) as an Aid for the Evaluation of Coronary Artery Plaques
 Near-Infrared Intravascular Ultrasound Coronary Imaging
 Near Infrared IVUS Imaging System
 Near-Infrared Spectroscopy (NIRS)
 NIR Spectroscopy
 TVC Imaging System

The use of specific product names is illustrative only. It is not intended to be a recommendation of one product over another, and is not intended to represent a complete listing of all products available.

Document History

Status	Date	Action
Reviewed	08/07/2025	Medical Policy & Technology Assessment Committee (MPTAC) review. Revised Description, Rationale and References sections.
Reviewed	08/08/2024	MPTAC review. Revised Rationale and References section.
Reviewed	08/10/2023	MPTAC review. Updated Rationale and References sections.
Reviewed	08/11/2022	MPTAC review. Rationale and References sections updated.
Reviewed	08/12/2021	MPTAC review. Rationale, Definitions, References, and Index sections updated.
Reviewed	08/13/2020	MPTAC review. Rationale and References sections updated.
	12/31/2019	Updated Coding section with 01/01/2020 CPT changes; added 93799 replacing 0205T deleted 12/31/2019.
Reviewed	08/22/2019	MPTAC review. Rationale and References sections updated.
Reviewed	09/13/2018	MPTAC review. Rationale, References, and Index sections updated.
Reviewed	11/02/2017	MPTAC review. Description/Scope, Rationale, Background/Overview and References sections updated. The document header wording updated from "Current Effective Date" to "Publish Date."
Reviewed	11/03/2016	MPTAC review. Rationale, Definition and References sections updated.
Reviewed	11/05/2015	MPTAC review. Rationale and Reference sections updated. Removed ICD-9 codes from Coding section.
Reviewed	11/13/2014	MPTAC review. Description, Rationale and References sections updated.
Reviewed	11/14/2013	MPTAC review. Rationale and References sections updated.
Reviewed	11/08/2012	MPTAC review. Rationale, References and Index sections updated.

Reviewed	11/17/2011	MPTAC review. Description, Rationale, Background, and References updated.
Revised	11/18/2010	MPTAC review. Title, Description, Rationale, Background, References, and Index updated. Clarified existing position statement and added a position statement addressing near-infrared intravascular ultrasound coronary imaging.
Reviewed	11/19/2009	MPTAC review. Description, Rationale, Background, References, and Definitions sections updated. Updated Coding section with 01/01/2010 CPT changes.
New	11/20/2008	MPTAC review. Initial document development.

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